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Robotic-assisted Skin sparing mastectomy and immediate reconstruction using latissimus dorsi flap a new effective and safe technique: A Comparative study

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Short title : Traditional versus robotic assisted latissimus dorsi flap – A comparative study

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ABSTRACT:

INTRODUCTION: Breast reconstruction is an essential part of breast cancer treatment. After skin sparing mastectomy, Immediate Breast Reconstruction (IBR) can be achieved using breast implants, autologous flaps (i.e. latissimus dorsi-myo-cutaneous flap (LDF)) or an association of both. Robotic assistance has gained popularity in many surgical fields including breast surgery. This study aims to compare the post-operative results of Robotic Assisted Latissimus Dorsi Flap (RALDF) to Traditional Latissimus Dorsi Flap (TLDF) for IBR after Skin Sparing Mastectomy (SSM) without nipple conservation.

MATERIALS AND METHODS: Between March 2016 and June 2019, all patients who underwent a SSM and a concurrent IBR with a TLDF were retrospectively compared to patients who underwent SSM and a concurrent IBR with a RALDF. Outcomes compared included operative time, length of hospital stay and complications rate.

RESULTS: 105 cases of SSM with a LDF based IBR were included in the study. 46 patients underwent RALDF and 59 patients underwent TLDF. Mean operative time was longer in the RALDF group (290.5min versus 259.7min). In binary regression, the concomitant placement of breast implant was the only factor associated with an operative time exceeding 290 minutes (p=0.032). Univariate analysis showed no significant difference concerning the rate of complications (p=0.061). After logistic regression, RALDF was associated with a decreased rate of complications (p=0.042; OR 0.37; IC 95% (0.142-0.966)).

DISCUSSION: SSM with IBR using RALDF is an effective and safe technique. This technique is actually associated with a lower complication rate at the expense of a longer operative time.

INTRODUCTION:

In the last few years, robotic assisted surgery has gained popularity. The use of robotic assistance in breast aesthetic and breast reconstructive surgery after breast cancer treatment is very recent [1].

On the basis of current evidence and the guidelines of the French National Cancer Institute (Institut National de Cancer), Nipple Sparing Mastectomy (NSM) is the Gold Standard approach in cases of prophylactic mastectomy [2]. Due to the limited data on the oncologic outcome after preservation of the Nipple-Areola Complex (NAC), this technique is still under evaluation in cases of therapeutic

mastectomies. On the other hand, Skin Sparing Mastectomy (SSM) without nipple sparing has been well documented and the absence of a negative oncologic outcome when compared to traditional non skin sparing mastectomy was proven.

After NSM and SSM, to achieve the best esthetic outcome, immediate breast reconstruction (IBR) could be proposed using either a breast implant, an autologous muscular or myo-cutaneous latissimus dorsi-flap (LDF) or a combination of both. Following the trend towards minimally invasive surgery, few articles reported the application of Non Robotic Laparoscopic-Assisted Breast Reconstruction with Latissimus Dorsi Muscle Flap [3-8]. When compared to traditional surgery and non-robotic laparoscopic surgery, the advantages of robotic assisted breast reconstructive surgery are its minimally invasiveness associated with an enhanced surgical exposure. This results in scar minimization, lesser tissue traction, lesser tissue trauma and the minimization of skin necrosis. On the other side, the potential longer operative time and the lesser volume of the flap due to the absence of the skin paddle are the most reported disadvantages of the robotic assisted breast reconstructive surgery with LDF.

Very little data exist in literature on robotic assistance in breast conserving surgeries, mastectomies and breast reconstructive surgeries (i.e. RALDF: robotic latissimus-dorsi-flap) and most of the published studies contain small numbers of patients [9-14].

The aim of this study is to compare the outcomes of patients who underwent SSM with RALDF to those who underwent SSM with TLDF.

MATERIALS AND METHODS:

All patients who underwent a SSM with a concurrent IBR using a LDF between March 2016 and June 2019 were included in the study. Patients operated traditionally were compared to patients operated with robotic assistance. Both techniques were applicable for all patients when skin paddle is not required, and the choice of technique was left to the surgeon to decide. Patients were not randomized due to logistic reasons (i.e. availability of the robotic systems that are shared with other departments) and the fact that not all of the breast surgeons master the RALDF technique.

All patients were informed about the surgical technique and the risks associated to either techniques.

Data was retrieved from the Institutional Breast Cancer Database and the study protocol was approved

by our Institutional Ethical Committee: NCT02869607. The data reported in the current analysis also include the patient data reported in the earlier publications [14, 15].

Surgical Techniques: *We previously described the surgical technique for SSM with IBR using RALDF* [14, 15]

In summary,

a) Surgical technique for mastectomy and immediate breast reconstruction with a traditional latissimus dorsi flap

All patients first underwent SSM in a dorsal decubitus position, then LDF in a lateral decubitus position. Patients were marked preoperatively in a standing position. The anterior border of the latissimus dorsi and the inframammary fold were marked. A single incision around the NAC was performed for SSM. Dissection was carried out to separate the glandular tissue and the fat lobules from the dermis at the level of the Cooper ligaments (also known as the crêtes de Duret), aided by the cold light retractor. Once mastectomy was finished, the patient's position was changed from DD to LD.

A horizontal elliptic incision following the pre-operative markings was carried out to perform the myo-cutaneous latissimus dorsi-flap. After harvest, the flap was then rotated, placed and fixed in the mastectomy space. Two suction drains were placed in the LDF donor site before wound closure and another one was placed in the mastectomy space. A de-epithelialization was carried out leaving a circular skin paddle corresponding to the neo-Areola.

b) Surgical technique for mastectomy and immediate breast reconstruction with a robotic assisted latissimus dorsi flap

All patients first underwent SSM in a dorsal decubitus position, then LDF in a lateral decubitus position. Patients were marked preoperatively in a standing position. The anterior border of the latissimus dorsi and the inframammary fold were marked. A single incision around the NAC was performed for SSM. When the SSM was finished, the patient's position was changed from DD to LD. For the majority (84.8%: 39/46) of patients in this group, the LDF harvest was performed through the mastectomy incision. For the remaining patients, a second 6-7cm vertical incision was performed at the level of the mid-axillary line. Subcutaneous dissection prepared the space for the single port

(Gelpoint® device) and was carried out downwards to the level of the infra mammary fold where the third robotic port was placed.

The Gelpoint® path single site device was inserted through the incision. Two robotic ports and a third port dedicated for the Airseal® insufflation system were placed through the Gelpoint® path. The robotic assisted part was performed under low insufflation pressure (7 mm Hg). A 0° camera was introduced through the central robotic port, and depending on the side operated on, monopolar scissors and bipolar forceps were introduced through the upper or the lower robotic ports.

The robotic assisted part starts with the superficial dissection of the latissimus dorsi muscle. The dissection was carried out downward (6-8cm below the inferior mammary fold) and upwards to the level of the Latissimus dorsi tendon that was resected at its insertion. The following step was the deep dissection of the Latissimus dorsi muscle that was carried out downwards to the previous limit and upwards to the level of the latissimus dorsi pedicle. The harvesting of the LDF was started at the level of the posterior attachments, then inferiorly, gradually mobilizing the muscle. This was performed using monopolar scissors. Two suction drains were placed in the LD donor site and passed through the infra-centimetric port incision, and one drain was placed in the reconstructed breast space (Figure 1).

Data collection: Data collected included *patient's characteristics* (age, body mass index (BMI), smoking status, presence of diabetes, ASA score, breast size), *prior breast cancer treatment* (sentinel lymph node biopsy, axillary lymph node dissection (LND), neoadjuvant chemotherapy, prior breast radiation therapy), *the type of the breast cancer* (new primary or a local recurrence) and *the type of breast reconstruction* (LDF with or without a breast implant).

Surgery related collected data included: the surgical technique, the robotic system used (da Vinci Si® or Xi® surgical systems, Intuitive surgical, Sunnyvale, CA), the number of ports used, the skin incision type, the duration of anesthesia, the surgical time, the association of surgical procedures (breast implant, LDF, sentinel lymph node, axillary LND and contralateral breast surgery).

Duration of anesthesia was calculated from the induction time to the extubation. The surgical time included all the procedures performed from the first skin incision to the end of skin suturing including the time required to change the patient's position.

The Clavien-Dindo classification was used to stratify and analyze the complications [16], and we added the term "grade 0" to describe the patients that did not present any complications. The re-operation rate, the complications type and the length of hospital stay were also reported. Data was prospectively collected and then retrospectively analyzed.

Statistics:

Means, medians and a Confidence Interval of 95% (CI95) were used to report the main quantitative characteristics of the analysis. The Chi2 test was used to compare qualitative variables, and the Student's t-test or the Anova test were used to compare continuous variables. Binary logistic regression using the SPSS 16.0 was used to predict the odds ratios (OR).

RESULTS:

Throughout the study period, 105 patients underwent SSM with a concurrent IBR using a LDF: 46 patients were operated for RALDF and 59 were operated for traditional latissimus dorsi flap (TLDF).

Indications and Type of Reconstruction:

There was a statistical significance in the distribution of the types of reconstruction (p=0.001):

In the Traditional technique group, there were 2 cases of Latissimus dorsi-myo-cutaneous flap, 6 cases of LDF with concomitant breast implant placement, 50 cases of LDF with the associated/surrounding fatty tissue and 1 case of LDF with the associated fatty tissue and a concomitant breast implant placement.

In the robotic assisted group, there were there were 7 cases of Latissimus dorsi myocutaneous flap, 9 cases of LDF with concomitant breast implant placement, 23 cases of LDF with the associated/surrounding fatty tissue (autologous LDF) and 7 cases of autologous LDF and a concomitant breast implant placement.

In the traditional group, 81.4% of the patients presented a new primary breast cancer versus 89.1% in the robotic assisted group (p=0.206); The remaining 18.6% and 10.9% of the patients in the traditional group and the robotic assisted group presented for a locally recurrent breast cancer.

In the TLDF group there were 7 cases of Ductal Carcinoma in Situ (DCIS), 45 cases of Invasive Ductal Carcinoma (IDC), 6 cases of Invasive lobular carcinoma (ILC) and one case of "other

histologic type" breast cancer. In the RALDF, 10 patients had DCIS, 24 patients had IDC, 11 patients had ILC and one patient presented "other histological type" breast cancer.

No significant difference was found between the two groups concerning the history of a previous ipsilateral breast surgery (50.8% of TLDF patients versus 50% of RALDF patients, (p=0.544)), Body Mass Index (p = 0.068), breast size, smoking status and the presence of diabetes (Table 1).

The mean age of the patients was 49.5 years and 58.1 years in the TLDF and RALDF groups respectively (p=0.001) (Table 2).

Additional or Associated Treatment:

The rate of Neoadjuvant chemotherapy was significantly higher in the TLDF group (50.8% vs 19.6%; p = 0.001). The rate of patients who had a previous history of ipsilateral breast radiation therapy (neoadjuvant or a history of radiation therapy) was significantly higher in the TLDF group (66.1% vs 30.5%, p< 0.0001).

During the same operation, sentinel lymph node biopsy was performed in 22% of TLDF versus 43.5% of RALDF (p=0.016), while there was no significant difference in the rates of axillary lymph node dissection (p= 0.563).

Adjuvant chemotherapy rates were similar in both groups (p=0.354). An adjuvant hormonal therapy was administered in 64.4% of the TLDF group compared to 69.6% of the cases in the other group (p= 0.635) (Table 1).

Surgical time:

The mean anesthesia and surgical durations were longer in the RALDF group compared to the TLDF group. The median anesthesia and surgical durations were respectively 356.9 min and 290.5 min for RALDF patients versus 327.5 min and 259.7 min for TLDF patients (Table 2-3). The association of breast implant to LDF was significantly longer than LDF alone (p=0.006 and p=0.032 for anesthesia and surgical time respectively).

Concomitant axillary lymphadenectomy and the breast size did not affect the operative time.

After binary regression, the LDF associated with the placement of breast implant was the only factor associated with an operative time exceeding 290 min (p=0.032). Breast Implant placement multiplied by three the number of surgeries exceeding 290min (OR 3.019 IC 95% (1.103-8.262)). Concomitant

188 axillary lymphadenectomy, BMI<25 and the robotic assistance were not associated with an operative
189 time exceeding 290min (Table 4).
190 *Post-operative Data:*
191 Mean hospital stay was 3 to 5 days with no significant difference between the two groups (p= 0.089)
192 (Table 2). After subgroup analysis, concomitant breast implant placement was significantly associated
193 with a longer mean hospital stay (Table 3). However, after logistic regression, breast implant
194 placement was not associated to a hospital stay of more than 3 days. The only factor that was
195 associated with a hospital stay of more than 3 days was overweight (BMI>25) with an OR of 2.48 (p =
196 0.045) (Table 4).
197 *Complications:*
198 When comparing the rate of all complications combined after univariate analysis, there was no
199 significant difference between the two groups (p= 0.061), however, a significant difference was found
200 in the severity of complications (p=0.03) (Table 1). Dorsal complications only (excluding breast
201 complications) were 23and 12 dorsal seromas (39% and 26.1%: non-significant), 2 and 1 dorsal
202 bleeding, 0 and 2 infections on seroma with 1 and 2 reoperations for non-robotic and robotic
203 procedures, respectively.
204 No significant difference was found when comparing no complications (grade 0) and mild
205 complications (grade1) to severe complications (grade 2-3) in both groups (p = 0.06).
206 After logistic regression, robotic assisted surgery was the only factor associated with a decreased risk
207 of complications (p = 0.042; OR 0.37; IC 95% (0.142-0.966)). Overweight (BMI>25), concomitant
208 breast implant placement, concomitant axillary lymphadenectomy, previous history of ipsilateral
209 breast radiation therapy and neo adjuvant chemotherapy were not factors associated with an increased
210 risk of complications.
211 After binary regression, no factors were significantly associated with an increased risk of severe
212 complications (grade 2-3) (Table 4).
213 **DISCUSSION:**
214 For patients, the most important advantage of SSM and IBR with RALDF is the absence of a dorsal
215 scar with less dorsal pain (pain not documented in this study), and in the majority of the cases (84.8%)

the presence of only one scar (the peri-areolar incision). However, the absence of a dorsal scar and skin paddle does not allow the reconstruction of a new-NAC. Moreover, RALDF without skin flap has a smaller volume than the myo-cutaneous TLDF.

In univariate analysis, the surgical time was longer in RALDF. This difference was no more significant after binary regression, which implies that this difference is related to other factors.

Our experience showed a significant decrease in the anesthesia and the surgical durations as the surgeons progressed along their learning curves. In a previous study, we defined three periods of time, with P1 corresponding to the year 2016, P2 corresponding to the year 2017 and P3 corresponding to the year 2018. We noticed an improvement in the surgical time when passing from P1 to P3. In the third period of the learning curve, the mean surgical time was 254 minutes and the mean time for the robotic assisted part was 45-60 minutes [15].

No difference was found between the RALDF group and the TLDF group concerning the post-operative complications. Furthermore, we noticed a trend towards a decrease in the post-operative complications in the RALDF group (OR = 0.37). This can be explained by the enhanced surgical exposure resulting in the minimization of tissue traction and the resultant tissue trauma and skin necrosis. Clemens et al. published similar results and concluded that RALDF harvesting is an efficient technique with low complications rate (16.7% among 12 RALDF versus 37.5% among 64 TLDF) (12). In the systematic review of Agha et al [17] that compared NSM and SSM, NSM was associated with a higher complication rate (22.6% versus 14% respectively). In the NSM group 15% presented a partial or complete nipple necrosis while only 3.9% of the patients in the SSM presented skin flap necrosis.

A prospective multicenter study reported by Jagsi et al [18] shows that autologous reconstruction appears to yield a superior patient-reported satisfaction and lower risk of complications than implant placement among patients receiving post mastectomy radiation therapy (PMRT). Currently, none of the published studies compared the esthetic outcome between robotic assisted and traditional SSM. However two studies evaluating axillary endoscopic mastectomy showed that endoscopic surgery were associated with a better esthetic outcome [19, 20].

Robotic surgery is associated with many advantages when compared to traditional surgery such as a smaller incision, better surgical exposure, decreased tissue trauma and an enhanced viability of the LDF. Acquiring a good experience in robotic surgery (pelvic or breast surgery) is considered crucial, this can be accomplished while assisting more experienced surgeons especially when the operating room is equipped with a dual robot console.

In **conclusion**, SSM with IBR using a RALDF is a feasible and efficient technique that leaves no dorsal scar. It can be performed in a wide range of patients including patients who underwent radiation therapy or patients with large breasts. IBR using a RALDF is not associated with an increased rate of complications when compared to TLDF. It can even be considered as a technique that reduces post-operative complications at the expense of a longer operative time, knowing also that operative time can be increased with increased BMI patients.

ACKNOWLEDGMENTS:

Not applicable

DECLARATIONS:

Database collection and analysis were performed at the Paoli Calmette Institute.

This work was approved by our institutional review board (IPC—Comité d'Orientation Stratégique).

All procedures performed in this study involving human participants were done in accordance with the French ethical standards and with the 2008 Helsinki declaration.

All included patients provided written informed consent before surgery, including the use of their data for research.

CONFLICTS OF INTEREST:

Nothing to declare

CONTRIBUTIONS:

Study concepts: Gilles Houvenaeghel

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Tables Legends:

Table 1: Patient's characteristics

Table 2: Comparison of operative and post-operative data between robotic assisted and traditional procedures.

Table 3: Comparison of operative and post-operative data according to the type of reconstruction, breast cup-sizes and axillary lymph node procedures.

Table 4: Binary Regression analyzing surgical time, mean hospital stay and post-operative complications

Figure Legend

Figure 1: Pre-operative Patient positioning, Per-operative positioning and Post-Operative Result.


1


2


3

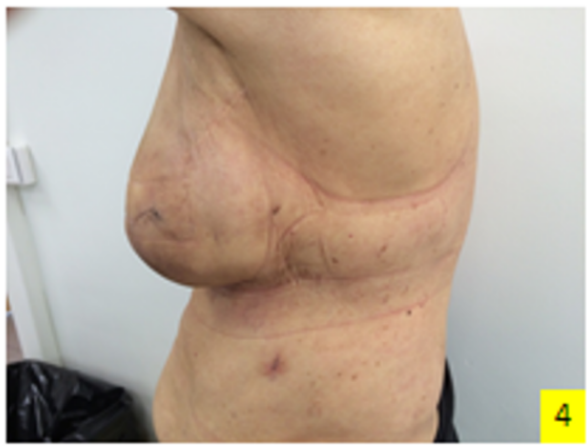

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Table 1: Patient's characteristics

		No Robotic		Robotic		Chi2
		N	%	N	%	p-value
		59	56.2	46	43.8	
Indication	Primary BC	48	81.4	41	89.1	0.206
	Local recurrence	11	18.6	5	10.9	
year of surgery	2016	19	32.2	10	21.7	0.507
	2017	18	30.5	18	39.1	
	2018	19	32.2	17	37.0	
	2019	3	5.1	1	2.2	
ASA	1	19	32.2	22	47.8	0.249
	2	39	66.1	23	50.0	
	3	1	1.7	1	2.2	
Smoking	No	47	79.7	36	78.3	0.634
	Yes	11	18.6	10	21.7	
Diabetes Mellitus	No	59	100	43	93.5	0.081
	Yes	0	0	3	6.5	
Cup size	A-B	29	49.2	19	41.3	0.725
	C	21	35.6	19	41.3	
	> C	9	15.3	8	17.4	
previous homolateral surgery	No	29	49.2	23	50.0	0.544
	Yes	30	50.8	23	50.0	
Type of IBR	LDF	2	3.4	7	15.2	**0.001**
	Implant + LDF	6	10.2	9	19.6	
	autologous LDF	50	84.7	23	50.0	
	Implant + A-LDF	1	1.7	7	15.2	
Histology	DCIS	7	11.9	10	21.7	0.076
	Ductal	45	76.3	24	52.2	
	Lobular	6	10.2	11	23.9	
	Others	1	1.7	1	2.2	
Endocrine therapy	No	21	35.6	14	30.4	0.365
	Yes	38	64.4	32	69.6	
SLNB	No	46	78.0	26	56.5	**0.016**
	Yes	13	22.0	20	43.5	
ALND	No	42	71.2	33	71.7	0.563
	Yes	17	28.8	13	28.3	
adjuvant chemotherapy	No	49	83.1	36	78.3	0.354
	Yes	10	16.9	10	21.7	
Neo adjuvant chemotherapy	No	29	49.2	37	80.4	**0.001**
	Yes	30	50.8	9	19.6	
Radiotherapy	No	18	30.5	25	54.3	**0.002**
	PMRT	2	3.4	7	15.2	
	previous RTH*	11	18.6	5	10.9	
	neo-adjuvant	28	47.5	9	19.6	

Complication	No	22	37.3	25	54.3	0.061
	Yes	37	62.7	21	45.7	
Grade complication	0	22	37.3	25	54.3	**0.03**
	1	34	57.6	14	30.4	
	2	0	0	2	4.3	
	3	3	5.1	5	10.9	
surgical incision	areolar	9	45.0	34	73.9	
	central elliptic	9	45.0	8	17.4	
	previous BCS incision	0	0	3	6.5	
	Inversed T incision	2	10.0	1	2.2	

Abbreviations: BC: breast cancer, BCS: breast conservative surgery, LDF: Latissimus dorsi-flap, A-LDF: autologous LDF (LDF with the associated fatty tissue), PMRT: post mastectomy radiation therapy, RTH: radiotherapy.

*Previous radiotherapy or neo adjuvant radiotherapy versus no radiotherapy or PMRT: $p<0.0001$.

Table 2: Comparison of operative and post-operative data between robotic and traditional procedures.

		Mean	Median	95% CI	p-value
age		53.3	52.0	50.6-56.0	
BMI		24.8	23.6	23.9-25.6	
Duration of surgery		273.3	270	261-286	
Duration of anesthesia		340.5	339	328-353	
Mastectomy weight		429	370	374-484	
Post operative stay		4.06	4.0	3.8-4.3	
age	no robotic	49.5	49	46.2-52.9	0.001
	robotic	58.1	59.5	54.0-62.2	
BMI	no robotic	24.1	23.4	23.1-25.1	0.068
	robotic	25.7	24.0	24.2-27.2	
Duration of anesthesia	no robotic	327.5	328	311-344	0.024
	robotic	356.9	346	337-377	
Duration of surgery	no robotic	259.7	266	244-276	0.016
	robotic	290.5	277	271-310	
Mastectomy weight	no robotic	405	388	349-460	0.329
	robotic	460	354	354-566	
Post operative stay	no robotic	3.86	4.0	3.55-4.18	0.089
	robotic	4.3	4.0	3.88-4.73	

Abbreviations: BMI: body mass index.

Table 3: Comparison of operative and post-operative data according to the type of reconstruction, breast cup-sizes and axillary lymph node procedures.

		Mean	Median	95% CI	p-value
Duration of anesthesia	LDF	304	316	264-344	**0.013**
	I + LDF	361.4	364	303-420	
	A-LDF	335	332	321-348	
	I + A-LDF	396.2	407	362-431	
Duration of surgery	LDF	244.6	248	207-282	0.081
	I + LDF	289.1	297	230-349	
	A-LDF	268.8	266	256-281	
	I + A-LDF	317.3	320	275-360	
Post operative stay	LDF	3.33	3.0	2.95-3.72	**<0.0001**
	I + LDF	4.47	5.0	3.81-5.12	
	A-LDF	3.85	4.0	3.56-4.13	
	I + A-LDF	6.0	6.0	5.0-7.0	
Duration of anesthesia	Cup size A-B	334.6	326	317-352	0.360
	Cup size C	344	340.5	324-365	
	Cup size >C	348.3	352	303-394	
Duration of surgery	Cup size A-B	267.9	266	252-283	0.506
	Cup size C	274.1	268	252-297	
	Cup size >C	286.4	293	244-329	
Duration of anesthesia	no ALND	337.4	332	322-353	0.449
	ALND	348.3	345.5	323-374	
Duration of surgery	no ALND	269.6	265	254-285	0.365
	ALND	282.4	280	259-306	

Abbreviations: LDF: Latissimus dorsi-flap, A-LDF: autologous LDF (LDF with the associated fatty tissue), I: implant, ALND: axillary lymph node dissection.
*LDF or A-LDF versus I+LDF or I+A-LDF: p=0.006
° LDF or A-LDF versus I+LDF or I+A-LDF: p=0.032

Table 4: Binary Regression analyzing surgical time, mean hospital stay and post-operative complications

	p-value	HR	CI 95%	
			Inferior	Superior
Duration of surgery <= vs > 290mn				
ALND vs no	0.628	1.253	0.503	3.119
Robotic LDF vs no	0.44	1.405	0.593	3.33
BMI >=25 vs <	0.152	1.852	0.797	4.308
I+LDF vs LDF	**0.032**	**3.019**	1.103	8.262
Post-operative stay <= vs > 3				
ALND vs no	0.563	1.316	0.519	3.334
Robotic LDF vs no	0.644	1.228	0.513	2.94
BMI >=25 vs <	**0.045**	**2.481**	1.02	6.036
I+LDF vs LDF	0.071	3.061	0.909	10.304
Complication				
BMI >=25 vs <	0.386	1.446	0.628	3.331
I+LDF vs LDF	0.666	1.259	0.442	3.587
Robotic LDF vs no	**0.042**	**0.37**	0.142	0.966
ALND vs no	0.132	2.313	0.777	6.885
previous RTH vs no	0.829	1.119	0.402	3.117
NACh vs no	0.252	0.493	0.147	1.653
Complication G2-3				
BMI >=25 vs <	0.863	0.879	0.206	3.762
I+LDF vs LDF	0.625	1.502	0.293	7.691
Robotic LDF vs no	0.206	3.306	0.517	21.133
ALND vs no	0.415	2.196	0.331	14.561
previous RTH vs no	0.69	0.695	0.116	4.165
NACh vs no	0.57	0.511	0.05	5.195

Abbreviations: BMI: body mass index, LDF: Latissimus dorsi-flap, A-LDF: autologous LDF (LDF with the associated fatty tissue), I: implant, ALND: axillary lymph node dissection, RTH: radiotherapy, NACh: neo-adjuvant chemotherapy.